Hein Zarni Aung

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EDUCATION

Stony Brook University **Expected Graduation: May 2027**

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Full Stack Development, Programming Abstractions, Systems Fundamentals I & II, Algorithms

INTERNSHIPS

Software Engineering Intern — Clustr **Jan 2026 – Sept 2026 (expected)**

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows, like/dislike interactions, and activity feed endpoints.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.

Software Engineering Intern — Prenora Dynamics LLC **Jun 2026 – Sept 2026 (expected)**

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, using the Supabase CLI to manage schema migrations and configure user auth/login flows.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University **Sep 2025 – Dec 2025**

Built a real-time SSL/TLS certificate monitoring pipeline (Python, Selenium) to detect fraudulent domains impersonating EZ-Pass tolling services.

CAMPUS INVOLVEMENT

Teaching Assistant – Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University **Aug 2025 – Dec 2025**

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

PROJECTS

TuneAcademy (React, NumPy, Essentia, Firestore)

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances, then pivoted to rule-based scoring algorithms under hackathon time and compute constraints.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy to Cloudflare Workers; broader app uses a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, and GCP hosting.

Discord Voice Channel Activity Monitor (Node.js, Discord.js, Cloud, PM2, AWS)

- Built a Node.js/Discord.js bot notifying users in real time when someone joins a voice channel, with customizable DM notifications, VIP lists, and per-server/per-user channel thresholds via JSON-based configuration.
- Deployed on a cloud server using PM2 and AWS; reached 50 concurrent active users.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: SQL, PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium